

Robotics Software Engineer

Innovative professional with experience in designing, developing, and deploying intelligent robotic systems using ROS2, Python, OpenCV, YOLOv8, Gazebo, and Unity.

Strong background in autonomous navigation, obstacle avoidance, sensor fusion, robot localisation, and human-robot interaction. Skilled in building AI-driven robotics solutions, integrating hardware components, and validating system performance through simulation and real-world testing. Demonstrates expertise in machine learning, computer vision, autonomous systems, and robotics research to deliver scalable, high-performance solutions. Adept at supporting cross-functional engineering initiatives in robotics, automation, and intelligent systems development. Recognised for combining technical precision, innovation, and problem-solving to improve operational efficiency and system reliability.

Areas of Expertise

- Robotics Software Development
- Autonomous Navigation Systems
- Technical Problem-Solving
- ROS / ROS2 Architecture
- System Testing & Validation
- Hardware-Software Integration
- Obstacle Detection & Avoidance
- Path Planning Algorithms
- Research-Led Innovation
- Data-Driven Optimisation
- Cross-Functional Collaboration
- Machine Learning Applications
- Scalable Robotics Solutions
- Human-Robot Interaction
- Algorithm Development

Education

MSc in Artificial Intelligence and Robotics | University of Hertfordshire, UK (Oct 2025)

Master of Computer Applications (MCA) | Kristu Jayanti College, India (July 2023)

Bachelor of Computer Applications (BCA) | Kristu Jayanti College, India (June 2021)

Technical Proficiencies

Programming: Python (PyTorch, TensorFlow, OpenCV), C++, SQL, MATLAB.

Robotics & AI: ROS 2, SLAM, MoveIt, Computer Vision, Generative AI, Reinforcement Learning.

Tools & Platforms: Linux/Ubuntu, Docker, Git, AWS (AI Fundamentals), Unity (Simulation).

Experience Highlights

Shift Manager | Greggs, London

Jan 2026 – Present

Direct end-to-end inventory analytics, logistics coordination, and waste control initiatives, enhancing operational efficiency and reducing resource losses. Execute targeted waste reduction strategies, strengthening sustainability practices whilst improving overall store efficiency. Integrate data-driven precision with strategic planning to deliver measurable business outcomes.

- Led team of 10+ staff in high-pressure environment, driving performance excellence and surpassing KPIs by 20%.
- Optimised inventory tracking and logistical processes, ensuring daily operations and improved stock accuracy.

MSc in Artificial Intelligence and Robotics | University of Hertfordshire, UK

Jan 2024 – Oct 2025

Gained advanced knowledge in artificial intelligence and robotics, with emphasis on intelligent systems design, machine learning, neural computation, robotics applications, and research-led problem solving. Built practical capability to design, implement, and evaluate AI-driven systems for industry and research environments. Key courses included Machine Learning, Neural Networks, Robotics Research, and Autonomous Systems.

- Strengthened expertise in machine learning, neural computation, artificial life with robotics, and AI theory and practice.
- Earned Young Indian Achiever 2025 distinction, highlighting exceptional performance and contributions within field.

Robotics Engineer (Internship) | Kristu Jayanti College – India

Mar 2023 – Jun 2023

Developed Voice-TTTS chatbot interface, facilitating seamless real-time communication and interaction with students and staff. Designed and executed industrial-grade simulations to evaluate and refine robot-human interaction protocols, ensuring operational safety and usability in public settings.

- Built end-to-end system on ROS2 architecture, demonstrating advanced capability in deploying scalable, industry-standard robotics frameworks.
- Integrated high-performance technology stack including Nvidia Jetson Nano, Rplidar, GPS, and temperature sensors, improving system responsiveness and environmental awareness.
- Enhanced user engagement by engineering specialised robotic system for university frontline assistance.
- Implemented ROS2-based navigation and leveraged OpenCV 3D mapping to enable accurate localisation, obstacle avoidance, and autonomous movement within complex environments.

Scientific Researcher (Internship) | ORBITX India Aerospace

Oct 2022 – May 2023

Collaborated on design and development of Guidance, Navigation, and Control (GNC) systems, enhancing flight stability and mission precision. Formulated and optimised reentry protocols, strengthening safety measures and ensuring structural resilience of reusable launch vehicle. Applied advanced mathematical modelling techniques to refine flight trajectories and maximise aerodynamic efficiency and system performance.

- Leveraged tools including Power BI, AutoCAD, Maya, Blender, and Google Colab to analyse complex datasets, improving system reliability by up to 21%.
- Conducted advanced research for Project Atal Yaan, driving innovation in prototyping, 3D modelling, and gizmo simulation to support vehicle development.

Robotics Engineer (Internship) | Mukkudal Robotics

Mar 2022 – Apr 2022

Developed system using ROS2 architecture, enabling efficient communication, control, and scalability of robotic functions. Delivered real-world solution aimed at reducing frontline exposure risks and improving operational efficiency for healthcare personnel during pandemic.

- Utilised specialised micro-robot for COVID-19 frontline assistance, supporting healthcare operations in high-risk environments.
- Integrated advanced hardware components, including GPS and temperature sensors alongside Voice-TTTS chatbot, enhancing real-time monitoring and user interaction.
- Designed and executed industrial-grade simulations to validate robot-human interaction protocols, ensuring safety, reliability, and practical deployment readiness.

Key Projects

AI-Enabled Autonomous Robot | Lead Developer: Designed and used YOLOv8-based navigation system using Python and ROS, improving obstacle detection and avoidance efficiency by 20%. Employed sensor-driven navigation workflows that enhanced real-time decision-making and autonomous movement accuracy. Simulated complex operational environments in Gazebo and Unity, validating sensor fusion algorithms before hardware deployment.

Project Atal Yaan – Reusable Launch Vehicle | Scientific Researcher: Supported development of Guidance, Navigation, and Control systems and reentry protocols for reusable launch vehicle, contributing to improved flight accuracy, mission stability, and recovery performance. Utilised advanced mathematical modelling to refine flight trajectories and reinforce structural integrity, strengthening aerodynamic efficiency and system resilience.

Certifications

Launching into Machine Learning (Google/Coursera)

Introduction to Generative AI (Google)

Technical Support Fundamentals (Google)